Haotian Liu

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EDUCATION

Xiamen University (XMU), Xiamen, China Sep. 2023 – Jun. 2027 (expected)
B.Eng. in Software Engineering | Overall Average: 82.66/100
Relevant Coursework:
Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); The Practice of Computer Programming (92).

PUBLICATIONS

PilotBench: A Benchmark for General Aviation Agents with Safety Constraints.
IJCNN 2026, **accepted**. Co-first author. arXiv:2604.08987
Long-Horizon Memory Governance for LLM Agents.
ICASSP 2027, under review. First author.
Calibrated Lexical Fusion for Noise-Robust Retrieval of Colloquial Chinese Procurement Queries.
ICASSP 2027, under review. Co-first author.

RESEARCH PROJECTS

DDRG: Reasoning-Graph Verification and Repair Apr. 2026 – Present
Research Collaborator | University of Tokyo
- Prototyped an inference-time reasoning-graph pipeline that samples candidate graphs, probes answer-sensitive claims, and gates local repairs conservatively.
- Evaluated Llama 3.1 70B on multiple reasoning benchmarks; case audits characterized parser, verifier, and repair-selection failures and informed conservative selection gates.

CETI: Evidence-Constrained Collaborative Task Interpretation Mar. 2026 – Present
First Author | Collaboration with ZGC Lab (PKU) and Waseda University
- Identified a cognitive gap–intent gap in repository-level AI coding and proposed CETI, which generates atomic task-constraint **Records** bound to repository evidence, enabling developers to inspect and revise an agent's interpretation *before* a patch is generated.
- Designed a four-stage pipeline (repository inspection, source binding, developer review, coding handoff) and a three-condition human study comparing FreeChat and one-/two-cycle Records on objective comprehension, patch correctness, and constraint satisfaction over ten FEA-Bench cases.

Long-Horizon Memory Governance for LLM Agents (LHMG) Dec. 2025 – May 2026
First Author | Remote Research Collaboration | School of Data Science, Fudan University
- Developed a memory-governance framework with conflict-aware revision graphs, No-Silent-Overwrite semantics, risk-gated memory conditioning, and governed forgetting to stabilize long-range LLM-agent reasoning.
- Observed a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory in controlled diagnostics; paper has undergone review at ICML before current ICASSP revision.

PilotBench: Safety-Critical Aviation Agents Dec. 2025 – Mar. 2026
Core Contributor | Remote Research Collaboration | CESS, Beihang University
- Co-developed PilotBench from 708 real trajectories, 34-channel telemetry, and nine flight phases; contributed to evaluations of 41 models.
- Contributed to Pilot-Score design and analyses identifying a precision–controllability dichotomy and a dynamic-complexity gap in high-workload phases.

INDUSTRY EXPERIENCE

Calibrated Lexical Fusion for Noise-Robust Retrieval of Colloquial Chinese Procurement Queries Nov. 2025 – Apr. 2026
AI Research Intern (ProcuraClaw system) | H World Group
- Built a deployed B2B procurement-search system for a global hotel group; designed a four-path hybrid retrieval pipeline (exact, fuzzy, synonym, and selective semantic via BGE-M3) fused with RRF, and a hotel-profile-driven quality function across five operational dimensions.
- Implemented online personalisation with Thompson Sampling and Posterior Dropout; achieved MRR 0.95, NDCG@5 0.91 (175 queries) with 6.2%/9.8% first-hit CTR gains in A/B tests; work written up as a paper under review at ICASSP 2027.

RESEARCH AND TECHNICAL SKILLS

AI and Programming: LLM agents and memory, RAG and information retrieval, multimodal learning, benchmark and evaluation design; Python, PyTorch, HuggingFace, Spring Boot, SQL, Java/JavaEE, C/C++, Unity.
Research and Academic Communication: Controlled experiment and human-study design, ablation and error analysis, academic writing, peer-review response, and research presentations; Git, Linux, Docker, and LaTeX.
Languages and Tests: Chinese (native); English (TOEFL iBT 90); Japanese (reading knowledge).